

Naive Bayes: The Intuition

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Classification problems ([Note 3](#)); Probability density function (PDF) ([Note 20](#)); Normal distribution ([Note 42](#)); Independent and mutually exclusive events ([Note 84](#)); Bayes' theorem ([Note 86](#)).

1. Overview

Key point: Naive Bayes computes, for each class, the probability of that class given the features, using Bayes' theorem, and predicts the class with the largest one. To make this possible, it assumes the features are independent within each class.

With conditional probability, independence and Bayes' theorem in place, we can build the **Naive Bayes classifier**. This Note develops the intuition on a tiny example; the next derives the mathematics in general, and the one after codes it.

2. The data

Key point: Eight cricket matches, three features (toss, venue, outlook) and the result. Predict the result of a new match.

The example is a small made-up dataset about one cricket team. Each match is one **observation** (one record, a row of the table). Each has three **features** (input variables, one column each) and one **target** (the output we predict):

match	toss	venue	outlook	result
1	won	Chennai	sunny	win
2	won	Mumbai	sunny	win
3	lost	Chennai	sunny	win
4	won	Mumbai	overcast	win
5	won	Chennai	sunny	win
6	lost	Mumbai	sunny	loss
7	lost	Chennai	rainy	loss
8	won	Mumbai	rainy	loss
new	lost	Mumbai	sunny	?

Figure 1: Eight matches and a new one to predict

- **toss:** did the team win or lose the toss?
- **venue:** Chennai or Mumbai.
- **outlook:** the weather, sunny, overcast or rainy.
- **result:** win or loss, the class to predict.

For a new match the toss is **lost**, the venue is **Mumbai** and the weather is **sunny**. Will the team win?

3. The strategy: compare posteriors

Key point: Compute $P(\text{win} \mid \text{features})$ and $P(\text{loss} \mid \text{features})$; predict whichever is larger.

Naive Bayes asks two questions:

$$P(\text{win} \mid \text{lost, Mumbai, sunny}) \quad \text{and} \quad P(\text{loss} \mid \text{lost, Mumbai, sunny})$$

The commas mean “and”: the three conditions hold together, so they could also be written with $\cap$. If the first probability is, say, 0.56 and the second 0.27, the classifier predicts a win. With more classes (for example win, loss, draw) the classifier computes one probability per class and again takes the largest.

4. Applying Bayes' theorem

Key point: Each posterior equals likelihood $\times$ prior / evidence. The evidence is the same for every class, so it can be dropped.

With A = the class and B = the three feature values, Bayes' theorem gives

$$P(\text{win} \mid \text{lost, Mumbai, sunny}) = \frac{P(\text{lost, Mumbai, sunny} \mid \text{win}) \times P(\text{win})}{P(\text{lost, Mumbai, sunny})}$$

and the same with “loss” in place of “win”.

The denominator, the evidence, is identical in both: it is just the probability of seeing these feature values, whichever the class. Dividing two numbers by the same positive amount does not change which is larger. So for choosing the class, only the **numerator** matters:

$$\text{score}(\text{class}) = P(\text{features} \mid \text{class}) \times P(\text{class})$$

The **priors** come straight from counting: 5 of the 8 matches were wins and 3 were losses.

$$P(\text{win}) = 5/8 \quad P(\text{loss}) = 3/8$$

5. The problem: specific combinations are rare

Key point: No winning match had exactly (lost, Mumbai, sunny), so the likelihood $P(\text{lost, Mumbai, sunny} \mid \text{win})$ would be 0.

The likelihood $P(\text{lost, Mumbai, sunny} \mid \text{win})$ asks: among the 5 winning matches, how many had all three conditions at once? None did. So it would be 0, and the score for “win” would be 0 however strongly the other evidence pointed to a win.

The rarity of exact matches is a general problem. A past observation that matches a new one on **every** feature at once is rare. With 3 features the search already failed here; with 10 or 20 features, almost every combination would never have been seen. The estimates would be 0 or based on one or two observations, and the classifier would be useless.

6. The naive assumption

Key point: Assume the features are independent within each class. Then the joint likelihood becomes a product of one-feature likelihoods, each estimated from many observations.

Naive Bayes makes a simplifying assumption: **within each class, the features are independent** of each other. From the independent events Note, the probability of independent events happening together is the product of their probabilities. So

$$P(\text{lost, Mumbai, sunny} \mid \text{win}) \approx P(\text{lost} \mid \text{win}) \times P(\text{Mumbai} \mid \text{win}) \times P(\text{sunny} \mid \text{win})$$

Each factor looks at one feature only, so it is estimated from all the observations of that class. The assumption is rarely exactly true, which is why the method is called **naive**, but it makes the estimates workable (ISL §4.4.4).

An everyday picture: to guess whether a new dish will taste good, we cannot wait for the exact same recipe to have been cooked before. Instead we judge each ingredient on its own record, and combine the verdicts. The next Note derives this step properly.

7. Computing the scores

Key point: Win: $5/8 \times 1/5 \times 2/5 \times 4/5 = 0.040$. Loss: $3/8 \times 2/3 \times 2/3 \times 1/3 = 0.056$. Predict loss.

Win (5 matches): the toss was lost in 1 of them, the venue was Mumbai in 2, and the weather was sunny in 4.

$$\text{score}(\text{win}) = \frac{5}{8} \times \frac{1}{5} \times \frac{2}{5} \times \frac{4}{5} = 0.040$$

Loss (3 matches): the toss was lost in 2, Mumbai in 2, sunny in 1.

$$\text{score}(\text{loss}) = \frac{3}{8} \times \frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} = 0.056$$

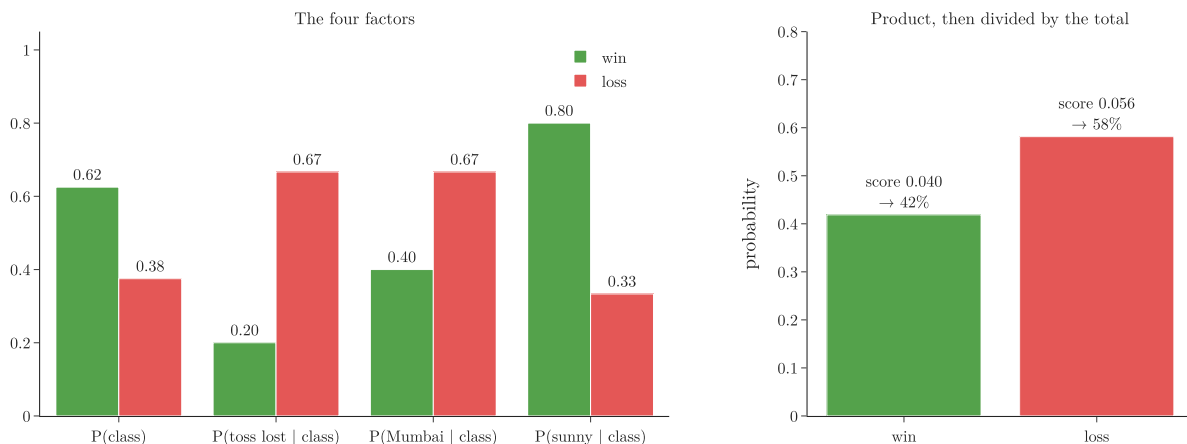


Figure 2: The four factors for each class, and the resulting probabilities

The loss score is larger, so Naive Bayes predicts a **loss**. Dividing each score by their sum turns them into probabilities: $0.040/0.096 = 0.42$, so 42% win and 58% loss (Figure 2, right).

Figure 2 (left) shows why: winning teams mostly won the toss and played in sunny weather, so “lost the toss” (0.20 against 0.67) and “Mumbai” (0.40 against 0.67) both point to a loss. Sunny weather points to a win (0.80 against 0.33), but not strongly enough to outweigh them.

8. Summary

Step	Win	Loss
Prior $P(\text{class})$	5/8	3/8
$P(\text{lost} \mid \text{class})$	1/5	2/3
$P(\text{Mumbai} \mid \text{class})$	2/5	2/3
$P(\text{sunny} \mid \text{class})$	4/5	1/3
Score (product)	0.040	0.056

- Naive Bayes picks the class with the largest $P(\text{class} \mid \text{features})$.
- The evidence is shared by all classes, so only likelihood $\times$ prior is compared.
- Exact combinations of feature values are rare; the naive independence assumption splits them into one-feature probabilities.

9. Sources

- **ISL:** James, G., Witten, D., Hastie, T. and Tibshirani, R. *An Introduction to Statistical Learning*, 2nd ed. Springer, 2021. Section 4.4.4, pp. 153–155.

10. Key terms

Term	Meaning
Naive Bayes classifier	A classifier that applies Bayes' theorem with the assumption that features are independent within each class
Naive assumption	The assumption that the features are conditionally independent given the class
Score	Likelihood $\times$ prior for a class; proportional to the posterior
Class prior	The share of training observations in a class